

Open Source Edge Platforms are Enriching IoT

The Internet of Things (IoT) is not just about connecting devices to the Internet, but also about gathering intelligence from the data at the edge, or near its source.

Today, more and more powerful IoT devices are coming to the edge. If you have storage, compute and network speed, all available at the edge close to the sensors, will you still send the data to the cloud? The answer is obviously, 'No!' We would choose to do the processing at the edge because it is faster. That's what is happening today.

Due to faster processing speeds, we are gradually moving to the edge, i.e., shifting from centralised to decentralised/distributed networks. It is the era of the blockchain, and soon

it is going to be the era of P2P (Peer to Peer) networks.

Edge computing

Edge computing is a method of optimising cloud computing systems by performing data processing at the edge of the network, near the source of the data. The source may be the sensors, actuators and devices close to you. All these generate a huge amount of data.

Today we have edge, fog and roof computing. All these are about the same thing — processing and computing closer to the device, but in different

ways, depending on the use case.

To understand the beauty of edge computing, let's consider a room that we want to make smart. Let's suppose you have a temperature sensor on your body and there are also temperature sensors in the room along with ACs. As soon as you enter the room, the sensors compare your body temperature with the room temperature and, accordingly, adjust the temperature of the ACs. This can be done by connecting the sensors to a Raspberry Pi board. But in the case of more than one type of sensor (pressure, humidity, etc), you can connect the processing device (such as

